Cheng (Kevin) Shao

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EDUCATION

University of Waterloo — Waterloo, ON
B.A.Sc. Electrical Engineering, Honours, Co-op — *Sep. 2025 – Expected May 2030*

EXPERIENCE

Electrical Site Coordinator Intern — Jul. 2026 – Present
Boralex — *Woodstock, ON*

- Monitored and oversaw electrical construction activities for the 125 MW/500 MWh Oxford Battery Energy Storage System (BESS) project.
- Audited contractor work against engineering plans to ensure strict adherence to quality control, safety guidelines, and environmental regulations.
- Drafted and reviewed progress reports, tracked procurement, and processed technical Requests for Information (RFIs) to maintain project schedules.
- Collaborated directly with the Site Manager and external contractors during site meetings to resolve technical deficiencies.

Electrical Team Member — Jan. 2026 – Present
University of Waterloo Baja SAE Team — *Waterloo, ON*

- Integrating and testing a Data Acquisition (DAQ) system using Arduino, MPU6050 accelerometers, and RPM sensors.
- Researching and coding strain gauges to extract live telemetry data, aiding in comprehensive system calibration.
- Fabricating space-efficient wire harnesses and designing circuit schematics for 2026 competition vehicles.

Project Team Member — Winter 2026
Waterloo Experience (WE) Accelerate Program — *Remote*

- Conducted accessibility and usability audits of five co-op community pages to identify UI/UX failures.
- Developed digital accessibility micro-course modules focusing on accessible HTML, CSS, and web design.
- Evaluated digital environments for strict AODA compliance and WCAG standard adherence.

PROJECTS

19V-13V Buck Converter Board (Baja SAE) | *KiCad, LM2596, Hardware Design Jan. 2026 – Present*

- Designed a custom step-down switching regulator PCB to drop a 19V drill battery input to a stable 13V output, reliably powering the vehicle's telemetry electronics.
- Engineered the schematic and routed the board using KiCad, incorporating an LM2596T-ADJ regulator, a 68μH (4A Isat) inductor, a 1N5822 Schottky diode, and Low-ESR capacitors for high-efficiency power conversion.

RPM Monitoring Device | *ESP32, C++, HW-477 Hall Effect Sensor* — *Jan. 2026 – Present*

- Engineered a custom rotational speed sensing solution using an ESP32 and an HW-477 Hall Effect sensor.
- Developed firmware to process analog magnetic signals and calculate accurate, real-time RPM telemetry.
- Integrated the sensor array into a broader vehicle DAQ framework to optimize off-road performance.

Smartwatch Sleep Monitor | *ECE 198 Design Project, Arduino, C++* — *Sep. 2025 – Dec. 2025*

- Prototyped a wearable Arduino UNO R4 and oximeter sleep tracker to address hospital-induced delirium.
- Programmed firmware to detect sleep cycles via heart rate variability thresholds (Start < 55, End > 65 BPM).
- Implemented IoT functionality to transmit automated daily sleep reports to caregivers via Wi-Fi.

TECHNICAL SKILLS

Programming/Firmware: C++, HTML, Arduino (Wiring)
Hardware & Tools: KiCad, ESP32, Arduino, Accelerometers (MPU6050), Multimeters, Soldering, Power Tools
Relevant Skills: PCB Design, Circuit Assembly, Wire Harnessing, Data Acquisition (DAQ), Technical Documentation
Spoken Languages: English (Fluent), Mandarin (Fluent Speaking/Listening, Intermediate Reading/Writing)